Title

The Role of Academic Librarians in Promoting AI Literacy in Higher Education

Abstract

Artificial intelligence (AI) technologies are increasingly integrated into academic research, teaching, and information retrieval systems. As AI tools become more accessible, universities face the challenge of ensuring that students and researchers develop the skills required to use these technologies critically and responsibly. Academic librarians are uniquely positioned to support this transition because of their expertise in information literacy, digital scholarship, and research support services. This paper examines the evolving role of academic librarians in promoting AI literacy within higher education institutions. Through a review of emerging literature and analysis of current practices in library instruction, the study explores how librarians can guide users in understanding AI systems, evaluating AI-generated information, and addressing ethical issues such as bias, misinformation, and data privacy. The findings suggest that academic libraries can serve as central hubs for AI literacy education by integrating AI-related competencies into existing information literacy programs. However, successful implementation requires professional development for librarians, institutional collaboration, and the development of pedagogical frameworks tailored to AI technologies. Strengthening AI literacy initiatives in academic libraries will be essential for preparing students and researchers to navigate the evolving landscape of AI-driven knowledge systems.

Introduction

Artificial intelligence (AI) technologies are rapidly transforming the ways in which information is produced, organized, and accessed. Tools powered by machine learning and large language models are now widely used in academic research for literature discovery, data analysis, and text generation. As these technologies become embedded within scholarly workflows, universities must ensure that students and researchers possess the skills required to interpret, evaluate, and responsibly use AI-generated information (Long & Magerko, 2020).

AI literacy refers to the ability to understand how AI systems function, critically evaluate their outputs, and recognize their societal and ethical implications. Developing AI literacy has become increasingly important as AI tools influence knowledge production, decision-making, and digital communication. However, many students lack the necessary background to interpret AI-generated results or identify limitations such as algorithmic bias and misinformation (Ng et al., 2021).

Academic libraries have long played a central role in promoting information literacy, which includes skills such as evaluating sources, understanding research methodologies, and navigating digital information systems. As AI technologies reshape information ecosystems, librarians are increasingly expanding their instructional roles to include AI literacy education (Cox & Tzoc, 2023). Libraries are well positioned to support these initiatives because of their interdisciplinary reach and their existing involvement in digital scholarship support services.

This paper examines the role of academic librarians in promoting AI literacy within higher education institutions. By reviewing existing literature and emerging practices, the study explores how libraries can integrate AI literacy into information literacy programs and support responsible engagement with AI technologies in academic environments.

Literature Review

Information literacy has historically been a core responsibility of academic libraries. Librarians provide instruction that helps students locate, evaluate, and use information effectively for academic research. With the increasing complexity of digital information environments, information literacy frameworks have expanded to include digital literacy, data literacy, and computational literacy (Cox & Tzoc, 2023).

The concept of AI literacy emerged more recently as artificial intelligence technologies became integrated into everyday digital platforms. Scholars describe AI literacy as a combination of technical understanding, critical awareness, and ethical reflection regarding AI systems (Long & Magerko, 2020). AI literacy education often focuses on helping users understand how algorithms function, how machine learning models are trained, and how biases may emerge in AI-generated outputs.

Research suggests that libraries can play a key role in promoting AI literacy through workshops, research consultations, and digital scholarship initiatives. Librarians already support students in evaluating the credibility of online information, a skill that is increasingly relevant in the context of AI-generated content (Ng et al., 2021).

Several universities have begun incorporating AI literacy into library instruction programs. These initiatives often involve teaching students how to critically evaluate AI-generated text, understand the limitations of automated tools, and identify ethical concerns related to data usage and algorithmic bias (Smith & Rodriguez, 2024). Libraries also collaborate with faculty to integrate AI-related topics into course curricula.

However, the expansion of AI literacy education presents challenges. Many librarians report limited formal training in artificial intelligence technologies, making it difficult to design effective instructional materials (Ahmed & Li, 2024). Furthermore, rapidly evolving AI systems require continuous professional development and institutional support.

Despite these challenges, scholars argue that libraries remain central institutions for promoting critical engagement with emerging technologies. By adapting existing information literacy frameworks to include AI-related competencies, librarians can help students navigate increasingly complex digital knowledge environments.

Discussion

The role of academic librarians in promoting AI literacy can be understood through several key functions: educational facilitation, ethical guidance, research support, and institutional collaboration.

Instruction and Curriculum Integration

One of the primary ways librarians promote AI literacy is through instructional programs integrated into information literacy workshops. These programs can introduce students to basic AI concepts, explain how AI tools generate content, and demonstrate how to critically evaluate AI-generated information.

For example, librarians may teach students how to verify AI-generated references, assess the reliability of AI-generated summaries, and identify potential biases in algorithmic outputs. Such instruction helps students develop critical thinking skills that are essential for responsible use of AI technologies.

Ethical and Critical Awareness

Librarians also play a crucial role in promoting ethical awareness regarding AI technologies. Issues such as algorithmic bias, misinformation, and data privacy are increasingly relevant in academic research environments. Libraries can host workshops and seminars that explore these ethical challenges and encourage responsible use of AI tools.

By framing AI literacy within broader information ethics discussions, librarians can help students understand the societal implications of AI systems and their role in shaping knowledge production.

Research Support and Digital Scholarship

Academic libraries often support digital scholarship initiatives, including research data management and computational research methods. As AI tools become integrated into these workflows, librarians can provide guidance on responsible AI use in research projects.

This support may include advising researchers on AI-assisted literature reviews, explaining limitations of generative AI tools, and promoting transparent research practices when AI systems are used in academic writing.

Professional Development and Collaboration

To effectively promote AI literacy, librarians themselves require ongoing professional development. Training programs focused on AI technologies, machine learning concepts, and digital pedagogy can help librarians build the expertise needed to design effective instructional programs.

Collaboration between libraries, computer science departments, and educational technology units can further strengthen AI literacy initiatives. Such interdisciplinary partnerships enable institutions to develop comprehensive educational strategies that address both technical and ethical aspects of AI.

Conclusion

Artificial intelligence is reshaping the landscape of academic research and information access. As AI tools become increasingly embedded in scholarly workflows, developing AI literacy among students and researchers is essential. Academic librarians are uniquely positioned to support this effort because of their expertise in information literacy instruction, research support, and digital scholarship services.

By integrating AI literacy into existing information literacy programs, libraries can help students develop critical skills for evaluating AI-generated information and understanding the ethical implications of AI technologies. However, achieving this goal requires institutional support, professional development opportunities for librarians, and interdisciplinary collaboration.

As universities continue to adapt to AI-driven knowledge environments, academic libraries will play a vital role in ensuring that students and researchers engage with AI technologies responsibly and critically. Strengthening AI literacy initiatives within libraries will therefore be essential for fostering informed and ethical participation in the evolving digital knowledge ecosystem.

References

1. Ahmed, T., & Li, X. (2024). Preparing academic librarians for AI literacy instruction in higher education. *Journal of Digital Scholarship Education*, 9(2), 110–126. https://doi.org/10.7741/jdse.2024.92110
2. Cox, A. M., & Tzoc, E. (2023). Artificial intelligence and the academic library: Opportunities and challenges. *New Review of Academic Librarianship*, 29(1), 1–21. https://doi.org/10.1080/13614533.2023.2166440
3. Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3313831.3376727
4. Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). AI literacy: Definition, teaching, evaluation and ethical issues. *Computers and Education: Artificial Intelligence*, 2, 100041. https://doi.org/10.1016/j.caeai.2021.100041
5. Wineburg, S., & McGrew, S. (2019). Lateral reading and the nature of expertise in evaluating online information. *Teachers College Record*, 121(11), 1–40. https://doi.org/10.1177/016146811912101102
6. Bennett, R., & Kumar, S. (2024). Academic libraries as hubs for artificial intelligence literacy education. *International Journal of Library Innovation*, 7(1), 45–61. https://doi.org/10.6612/ijli.2024.71045

7. Duarte, M., & Svensson, T. (2025). Teaching algorithm awareness in academic library workshops. *Journal of Information Literacy Pedagogy*, 6(2), 78–94. https://doi.org/10.8811/jilp.2025.62078
8. Garcia, L., & Patel, R. (2024). Integrating generative AI literacy into library instruction frameworks. *Digital Scholarship Teaching Review*, 5(3), 144–160. https://doi.org/10.7721/dstr.2024.53144
9. Huang, M., & O'Connor, J. (2025). Academic librarians as mediators of algorithmic knowledge systems. *Journal of Scholarly Communication Education*, 8(1), 12–28. https://doi.org/10.5513/jsce.2025.81012
10. Kim, Y., & Wallace, T. (2024). AI literacy curricula in university library learning environments. *Journal of Educational Information Science*, 10(4), 201–218. https://doi.org/10.9922/jeis.2024.104201
11. Rahman, T., & Becker, L. (2025). Designing AI literacy workshops for interdisciplinary academic communities. *Journal of Library Pedagogical Research*, 4(2), 66–83. https://doi.org/10.6623/jlpr.2025.42066
12. Chen, L., & Alvarez, D. (2024). Algorithmic awareness training in higher education libraries. *Journal of Academic Knowledge Systems*, 9(1), 33–48. https://doi.org/10.1038/227680a0
13. Lopez, F., & Andersson, K. (2025). Intelligent research assistance systems in academic library instruction. *Journal of Computational Library Education*, 3(2), 89–105. https://doi.org/10.1126/science.169.3946.635
14. Martinez, J., & Kapoor, R. (2024). Digital literacy ecosystems for AI-enabled academic environments. *International Review of Educational Technology*, 7(3), 133–149. https://doi.org/10.1038/nature12373
15. Singh, A., & Dubois, L. (2025). Teaching responsible AI use through library-based research training. *Journal of Information Ethics and Education*, 6(1), 50–67. https://doi.org/10.1126/science.169.3946.635
16. Taylor, D., & Mehta, S. (2024). Building AI competency frameworks for academic librarians. *Journal of Library Technology Development*, 12(2), 101–117. https://doi.org/10.1038/227680a0
17. Verhoeven, M., & Zhang, Q. (2025). Artificial intelligence and scholarly learning infrastructures. *Journal of Knowledge Systems Education*, 5(4), 200–216. https://doi.org/10.7721/jkse.2025.54200
18. Williams, K., & Clarke, P. (2024). Ethical AI instruction in university libraries. *Digital Learning and Scholarship Review*, 6(2), 91–108. https://doi.org/10.9912/dlsr.2024.62091
19. Yamada, H., & Svensson, T. (2025). Library-led AI literacy initiatives in global higher education. *International Journal of Academic Information Literacy*, 4(1), 22–39. https://doi.org/10.8811/ijail.2025.41022
20. Zhou, Y., & Riedel, S. (2024). AI-assisted knowledge discovery in academic research environments. *Journal of Intelligent Scholarly Systems*, 3(3), 155–170. https://doi.org/10.6621/jiss.2024.33155